

SMART DISPLAY MODULE SPECIFICATION

1.0 Inch Smart Knob Display	
Model:	UEDX12115010-MD33E
Version:	V1.0
Date:	2025-04-14

Customer Confirmation

Approved by	Notes

REVISION HISTORY

[illegible]

TABLE of CONTENTS

1. INTRODUCTION	4
1.1 Features	4
1.2 Appearance picture	5
2. PRODUCT INFORMATION	6
2.1 Pin Overview	6
2.2 Display Information	7
2.3 Voltage & Current	7
2.4 Reliability Test	8
2.5 software support	8
IDE:	8
Display graphics library: LVGL	8
3. SCHEMATIC	9
4. MECHANICAL DRAWING	10
5. DETAILED INFORMATION	11

1. Introduction

1.1 Features

Brief Info:

- 1) Outline Dimension: ϕ 33 Round
- 2) Interaction Method: Rotate and Press
- 3) Shell Color: Black/White/Silver/Customized
- 4) Power: DC 5V, 1A

MCU

- 1) OS: FreeRTOS
- 2) CPU: ESP32-C2 120Mhz
- 3) Flash: 4 MB
- 4) Interface: UART/USB
- 5) Support 2.4GHz Wi-Fi、BLE 5、BLE Mesh

For more information on ESP32-C2, please refer to the following link:[datasheet_en.pdf](#)(English) or [datasheet_cn.pdf](#)(Chinese)

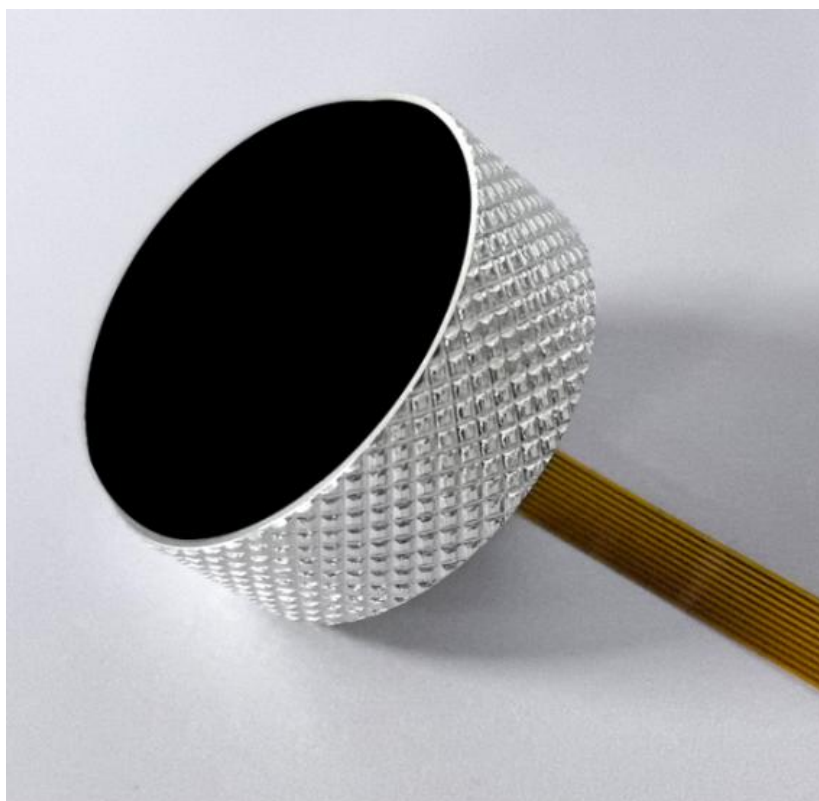
Display

- 1) Size: 1.02 Inch
- 2) Resolution: 128*115
- 3) Mode: IPS
- 4) Interface: 4SPI
- 5) Brightness: 400 cd/m²
- 6) Touch: without

Other

- 1) Operation Temperature: -20~60°C
- 2) Storage Temperature: -30~70°C

1.2 Appearance picture



2. Product information

2.1 Pin Overview

Display:

IPS Display Pin	ESP32C2 Pin
LEDK	GPIO8
LEDA	VDD 3.3V
DC	GPIO3
CS	GPIO2
SCL	GPIO1
SDA	GPIO0
RST	GPIO10

Note: LEDK is LED Cathode, controlled by GPIO8

Encoder:

Encoder pin	ESP32C2 Pin
PHA	GPIO5
PHB	GPIO6

Button:

Button pin	ESP32C2 Pin
SWKEY	IO9

10pin FPC:

FPC number	adapter pin	ESP32C2 Pin
1	5V	5V
2	PB7	GPIO4
3	GND	GND
4	RX2	GPIO18
5	TX2	GPIO7
6	RX1	RXD
7	TX1	TXD
8	-	CHIP-EN
9	SK & D+	-
10	SD & D-	-

Note: The FPC (Flexible Printed Circuit) is the flexible cable extending from the rotary knob display, while the adapter board is the circuit board used to connect the display for power supply and firmware programming/burning.

2.2 Display Information

Features	Details	Unit
Display Size(Diagonal)	1.02	inch
LCD type	α -Si TFT	-
Display Mode	IPS/ Transmissive / Normally Black	-
Resolution	128RGBx115	-
Active Area	24.36(H) $\times$ 21.89(V)	mm
Module Outline	26.71(H) $\times$ 26.22(V) $\times$ 1.86(T)	mm
Display Colors	262K	-
Interface	4SPI	-
Driver IC	GC9107	-
TP Viewing Area TP	24.76(H) $\times$ 22.29(V)	mm
TP Outline(assembly)TP	31.8(H) $\times$ 31.8(V) $\times$ 1.1(T)	mm
Luminance on surface	400	cd/m ²
View Direction	All	Best image
Contrast ratio	1200:1	
Color gamut	70%	
PPI	133	-
Operating Temperature	-20 $\sim$ 60	°C
Storage Temperature	-30 $\sim$ 70	°C

2.3 Voltage & Current

Item	Conditions	Min	Typ	Max	Unit
Power Voltage	DC	4.0	5.0	5.5	V
Operation Current	VCC= +5V, Maximum backlight current	50	100	150	mA
	VCC= +5V,backlight off	-	50	-	mA
Recommended power supply:5V 1A DC					

2.4 Reliability Test

Item	Conditions	Min	Typ	Max	Unit
Working Temperature	60%RH at 5V voltage	-20	25	70	C
Storage Temperature	---	-30	25	85	C
Working Humidity	25°C	10%	60%	90%	RH
ESD	---	Contact: ±4KV Air: ±8KV			KV

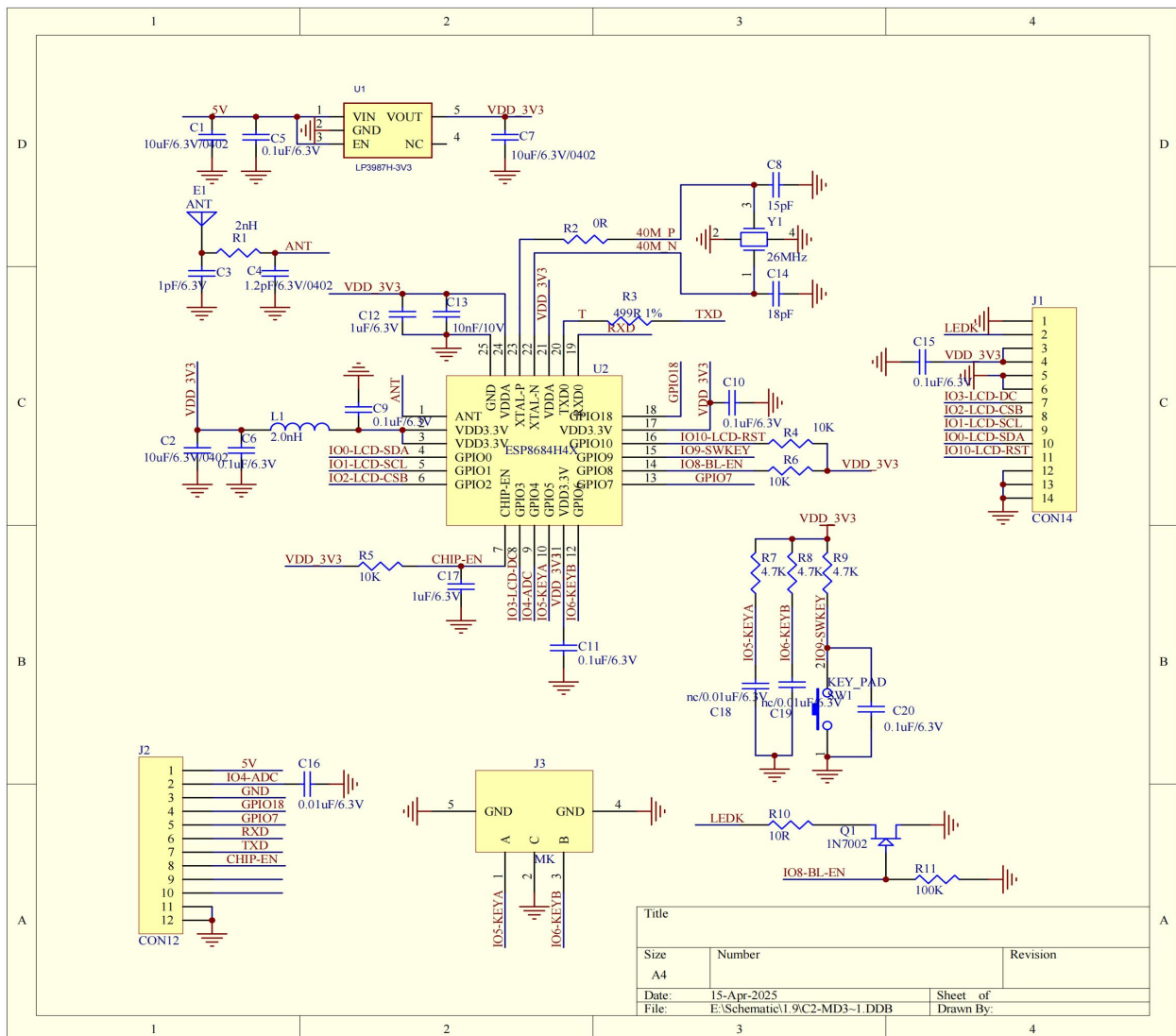
2.5 software support

IDE:

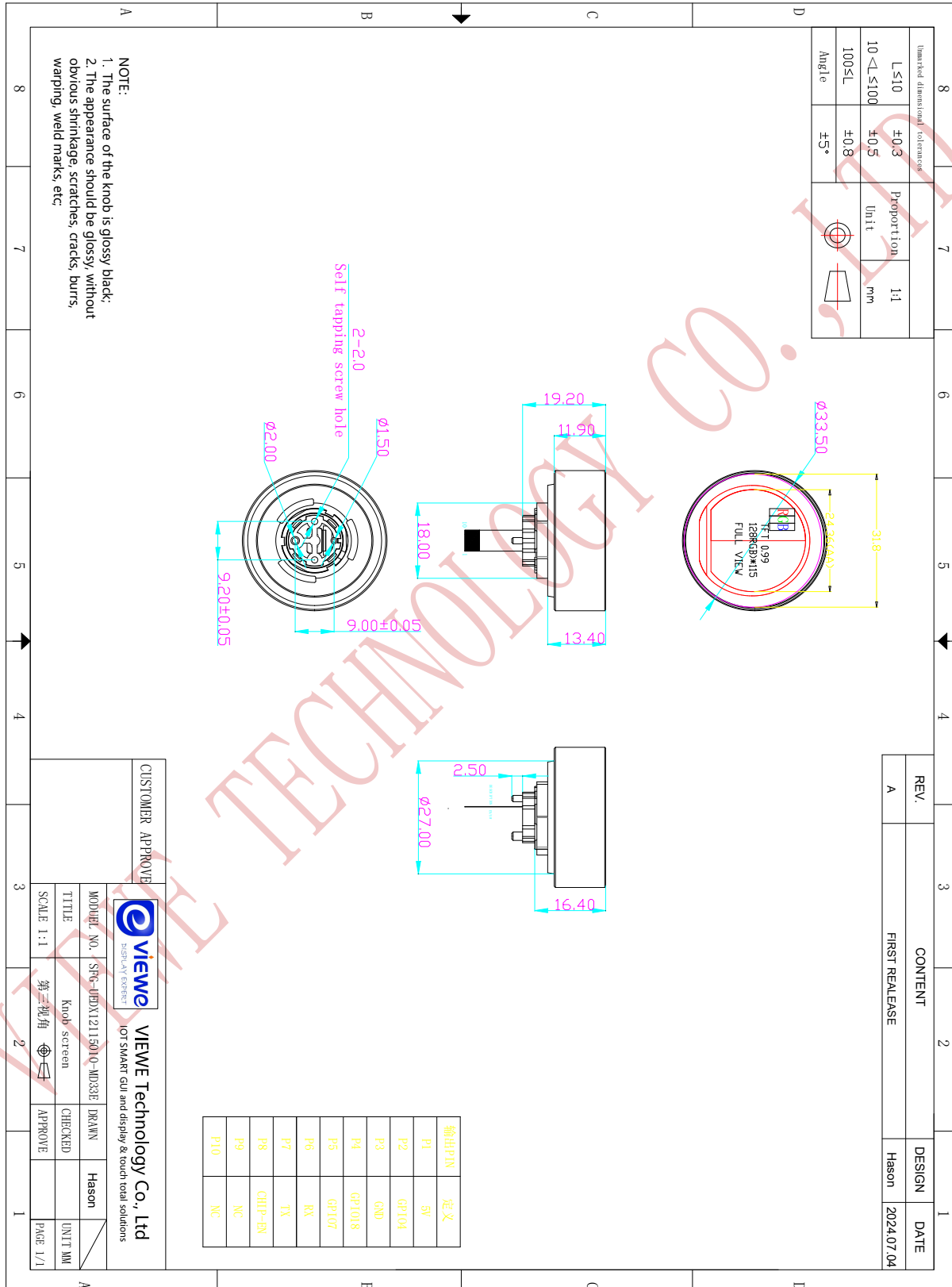
Software name	Version	Development environment configuration link
Arduino IDE	3.1.0(esp32)	
ESP-IDF	5.4	
PlatformIO IDE		

Display graphics library: LVGL

3. Schematic



4. MECHANICAL DRAWING



5. Detailed information

GitHub: <https://github.com/VIEWESMART/UEDX12115010-MD33E-1inch-Knob>

VIEWE TECHNOLOGY CO., LTD